

## 8.2 Protocols

### Network protocols

#### Introduction

You have seen how networks connect computers together. Networks need hardware, software and connectivity. Networks also need protocols. Protocols are standards for communication. In this section you will learn why protocols are important. Two important network protocols are Ethernet and Wi-Fi. There are many different protocols in use nowadays. Protocols are grouped together into families of protocols. One family of protocols is responsible for one part of the overall communication process.

### What are standards?

Computer communications rely on common standards. Standards are common ways of doing things. If two computers use the same standards then they are able to send messages to each other. There are standards for the way hardware is designed, and for the way that equipment is linked together. ASCII and Unicode are standard ways of coding characters as binary numbers.

Some standards are very popular and widespread. Manufacturers decide to make equipment and software to match these standards. That means the computers they produce can communicate with other computers. It makes the manufacturers' products more attractive. It is in those companies' commercial interest to use standards.

Some standards are regulated by organisations. The organisations will test equipment or software to make sure it matches the standard. If it does conform to the standard then the manufacturer is allowed to state this in the product description. Manufacturers may be allowed to use the trademark or symbol. This makes their product more successful. For this reason, manufacturers work hard to meet the latest standards.

### What are protocols?

Protocols are standards for the way signals are sent over networks. Protocols are rules about how signals are prepared, sent and received. There are many different communication protocols. We will learn about them in this section.

### Ethernet

Ethernet is a system of protocols for preparing and transmitting data in a wired network. In the Ethernet system communications are broken up into small sections called packets. There are rules for the binary signals that mark the start and end of a packet. A packet is sent together with an address. The packet travels down the network until it finds the computer with the right address. It is then copied into that computer's memory.

Ethernet is supported by a non-profit making organisation called the Ethernet Alliance. This organisation checks equipment and promotes the Ethernet system.



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## Wi-Fi

Wi-Fi is a protocol used for wireless networking. Like Ethernet, it is a family rather than a single protocol. A wireless local area network is called a WLAN. Wi-Fi is a protocol used by WLANs.

Devices that are Wi-Fi enabled have been built to use the Wi-Fi protocol. These devices normally have a sticker on them to confirm they are Wi-Fi compatible. A device that can use the Wi-Fi protocol can connect to any Wi-Fi network. A place where a Wi-Fi network is in place is called a Wi-Fi hotspot.

You may recognise the Wi-Fi logo.

For example, if your tablet or phone is Wi-Fi certified, you will be able to pick up Wi-Fi connections. If you know the password to a Wi-Fi network, you will be able to log in to the network. Many businesses such as coffee shops offer a free Wi-Fi connection.

Wi-Fi is a trademark of a non-profit organisation called the Wi-Fi Alliance. This organisation tests devices to make sure they are compatible with the Wi-Fi protocol. Only product that pass the tests can call themselves Wi-Fi certified.



### Test yourself

1. What is a WLAN? Which family of protocols is designed for a WLAN?
2. If a product has a sticker on it that says Wi-Fi certified. What does that mean?
3. What types of network use the Ethernet standard?
4. What is a Wi-Fi hotspot?

### Learning activity

Find examples in places near you of networks using the Ethernet or Wi-Fi protocols. If your school or college has a network ask the network technician to tell you about the protocols it uses.

Are there Wi-Fi hotspots near to or inside your school or college? Consider a class project that involves mapping out networks in your area.

### Extension activity

You have learned about the Wi-Fi and Ethernet protocols. Another protocol that is available is called Bluetooth. What is the Bluetooth protocol and what is it used for?



### 3.5 Fundamentals of computer networks

Describe the four-layer TCP/IP model:

- application layer
- transport layer
- network layer
- link layer.

Explain the purpose and use of common network protocols including:

- TCP (Transmission Control Protocol)
- HTTP (Hypertext Transfer Protocol)
- HTTPS (Hypertext Transfer Protocol Secure)
- FTP (File Transfer Protocol).

#### Additional information

Students should know what each protocol is used for.

### HTML or HTTP?

Be careful not to confuse HTML and HTTP.

HTML: Hypertext markup language is the language we use to define a web page. Your browser can read HTML. The HTML tells your browser what the page looks like. The browser shows you the page on screen.

HTTP: Hypertext transfer protocol is the protocol that gets this web page information to your browser from the remote server.

# The four-layer model

## Introduction

You have learned that protocols are standards for communication. Shared protocols allow computers to communicate with each other. In this section and the next you will learn about the protocols that allow the modern Internet to function.

## DARPA and the Internet

TCP/IP is a group of standards and protocols for high speed links over a wide area network (WAN). TCP/IP was developed in 1969 by a US government agency called DARPA. Since 1969 this system of links has grown into a worldwide network system known as the Internet.

There are a lot of protocols that control computer communications. These shared protocols are what make the Internet work. The protocols can be organised into a four-layer model. This is sometimes called the TCP/IP model or the DARPA model. The four layers are:

- application layer
- transport layer
- network layer
- link layer.

In this section you will learn about the application layer. On pages 236–237 you will learn about the other three layers.

## Application layer

The application layer contains the protocols that make common Internet applications work. This includes the web browser software that you use to look at websites. It also includes applications that let you send and receive emails.

The most widely used application layer protocols include:

- hypertext transfer protocol (HTTP)
- hypertext transfer protocol – secure (HTTPS)
- file transfer protocol (FTP)
- simple mail transfer protocol (SMTP)
- Internet message access protocol (IMAP)

## Hypertext transfer protocol (HTTP)

HTTP is the protocol used for transmitting web pages between a server and the web browser. HTTP is the protocol that defines “hypertext”. Hyperlink means text in a website that makes a link to other web pages. Your web browser understands this protocol. When you click on a link your browser connects to a new web page. By clicking on links you can move around the Internet looking at different websites.

If you look at the name of a website in your browser you will see it begins:

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http://

This shows that the web page uses the hypertext transfer protocol.

## HTTP secure (HTTPS)

Another version of HTTP is called HTTPS. The “S” stands for secure. HTTPS is a safer way of connecting to a web page. Using HTTPS stops the people you don’t want to read your messages from doing that. It can also stop people tricking you into giving them your personal details or money.

HTTPS includes extra security features such as encryption. Encryption means your messages to an HTTPS site are put into a code before they are sent over an Internet connection.

You can tell which websites you visit are protected by HTTPS. The name of the web page begins with the name of the protocol:

https://

## File Transfer Protocol (FTP)

FTP stands for file transfer protocol. FTP is a way of getting data files from one computer to another. The computer that has the files is called the server. The computer that receives the files is called the client.

FTP is an older protocol than HTTP. FTP lacks many modern Internet features.

## Simple mail transfer protocol (SMTP)

Simple mail transfer protocol (SMTP) has been used for many years. It is the protocol used to send email messages and attach files to emails. SMTP is a delivery protocol. That means SMTP delivers an email message to an address. It does not fetch your email messages for you. Originally, SMTP was text only but nowadays emails can include all types of images and file attachments.

## Internet message access protocol (IMAP)

The Internet message access protocol (IMAP) is a protocol that gets your emails. Companies that offer email services store the emails on a large computer called a mail server. All emails sent to you will be stored in an inbox on the mail server. IMAP lets you look at the e-mail messages in your inbox, using an Internet connection. You can see the headings of the emails and open any of them to read the message.

Many email applications such as Gmail and Outlook use IMAP.

### Test yourself

1. How can you tell whether a website uses the HTTP protocol or the HTTPS protocol?
2. What extra features are included in the HTTPS protocol?
3. A web page contains hyperlinks. What are hyperlinks?
4. SMTP and IMAP work together to make email systems work. Explain how the work of sending and reading emails is divided between the two protocols.

### Learning activity

Investigate a range of Internet sites. Which sites use HTTP? Which use HTTPS? What type of site uses HTTPS? Write a report on what you have found.

### Extension activity

Websites are read using software called a web browser. What are the most common web browsers in use today? See whether you can find out what percentage of web users use each type of browser.



## Syllabus reference

### 3.5 Fundamentals of computer networks

Describe the four-layer TCP/IP model:

- application layer
- transport layer
- network layer
- link layer.

Explain the purpose and use of common network protocols, including:

- transmission control protocol (TCP)
- user datagram protocol (UDP)
- Internet protocol (IP).

#### Additional information

Students should know what each protocol is used for.

# The four-layer model (continued)

## Introduction

You have learned about the protocols that make the application layer of the four-layer model. In this section you will learn about the other three layers of the model.

## Transport layer

A web host is a company that operates a computer connected to the Internet. The transport layer holds protocols that allows web hosts to send messages and share data. It is sometimes called the host-to-host transport layer. A message between two web servers may go through many short connecting links. The transport layer does not deal with these smaller links. It controls the communication at the two ends. Transport layer protocols are responsible for establishing the connection between the two servers. The web services of the application layer can only work because the transport layer takes care of sending messages between two computers. The transport layer ensures the two computers are using the same settings, such as language and the size of the data packets they send and receive.

Two key protocols of the transport layer are transmission control protocol (TCP) and user datagram protocol (UDP):

- TCP allows the computers at the two ends of the communication link to recognise each other. Using TCP, each computer can read the messages sent by the other computer. TCP emphasises accurate transmission not speed.
- UDP is an alternative to TCP. It has reduced error-checking and works more quickly. It is only suitable when a message is short and the connection is very reliable.

## Network layer

The transport layer protocols control the two ends of the transmission link. The network layer takes care of the detailed connections in between. A message must pass through many different connections on its route through the Internet. The network layer has protocols that control this journey. For this reason it is also sometimes called the Internet layer. We can imagine a packet of data being passed from point to point on a journey through the Internet. It may pass through many different smaller networks, servers and routers. But it will end up at the right destination. This process is controlled by network layer protocols.

The most important protocol in this layer is called the Internet protocol (IP). The IP makes sure that every message gets to the right Internet address.

Every web host has an IP address. Messages are split into packets and each packet is sent down an Internet route to the right IP address. Then the packets are put back together and the message is delivered.

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## Link layer

You have learned that TCP and IP work together to deliver messages over an Internet connection. The TCP/IP will deliver messages to any kind of computer or network that has an Internet connection. TCP/IP protocols are the same for all computer systems. That's why all different types of computer can use the Internet.

The final layer of the four-layer model is the link layer. The protocols in this layer convert the messages from the TCP/IP system into a form that the local network can use. Ethernet is a protocol at the link layer, which enables signals to go down local network connections. Device drivers convert signals from network hardware devices and transfer them to the processor. The network interface card converts binary signals from network devices into binary data inside each computer.

## Summary

There are four protocol layers. All four layers are involved when a signal is sent through a network system such as the Internet.

| Layer                            | Description                                                                                           | Protocols                            |
|----------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Application layer</b>         | This allows software applications such as web browsers and email programs to work.                    | HTTP<br>HTTPS<br>FTP<br>SMTP<br>IMAP |
| <b>Transport layer</b>           | This sends messages from one web host to another. It controls the two ends of the link.               | TCP<br>UDP                           |
| <b>Network or Internet layer</b> | This controls how a message travels through the Internet from one end of the link to another.         | IP                                   |
| <b>Link layer</b>                | This turns messages from the Internet into a form that works on the local network or computer system. | Ethernet                             |

These layers may be thought of as stacked on top of each other. A signal passes down through the stack. It starts at the application layer, then to the transport layer. Next the network layer puts it on a route, and the link layer converts it into locally transmissible form. When it arrives, the signal passes back up through the layers. It ends up at the application layer in a new location.

### Learning activity

Copy the summary table. Add an extra column to give a short description of each of the protocols in the table.

As a class project, turn this into a large poster for the computer room.

### Test yourself

1. What are the main differences between TCP and UDP? What is the advantage and disadvantage of each protocol?
2. The IP controls packets and addressing. Explain what packets are and what IP addresses are.
3. Why does the Internet need both the transport layer and the network layer? Explain how the two layers work together to give a complete Internet service.
4. How does a network interface card help with the work of the link layer?

### Extension activity

Find out more about the history of the Internet and the key figures involved in development of the protocols that enable the Internet to work. Write a magazine article for students.